

Waves

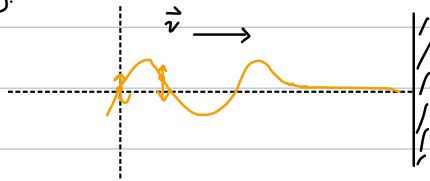
Three main types:

1. Mechanical waves (water, sound, seismic, ...)
2. Electromagnetic waves (visible light, radio, ...)
3. Matter waves (quantum mechanical view of fundamental particles)

A wave travelling along the direction of $\vec{v}$:

- Transverse wave ($\Delta \vec{y} \perp \vec{v}$)

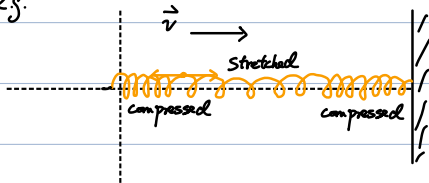
e.g.



a string attached to wall

- Longitudinal wave ($\Delta \vec{y} \parallel \vec{v}$)

e.g.



a spring attached to wall

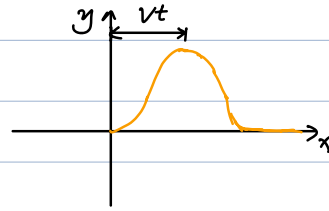
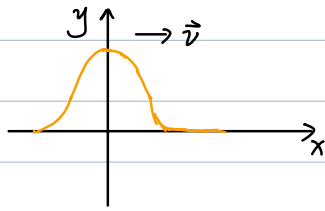
For mechanical waves, need

1. source of disturbance
2. medium
3. physical connection between adjacent portions of the medium.

Basic variables of periodic waves:

- Amplitude A
 - frequency f , angular frequency ω , period T
 - Wavelength λ : distance of points whose oscillations differ by 2π .
(periodic waves)
- } similar to simple harmonic motion

1D travelling waves



Displacement y is a function of both x & t :

$$y = f(x, t)$$

Right-moving pulse:

$$y = f(x - vt)$$

Left-moving pulse:

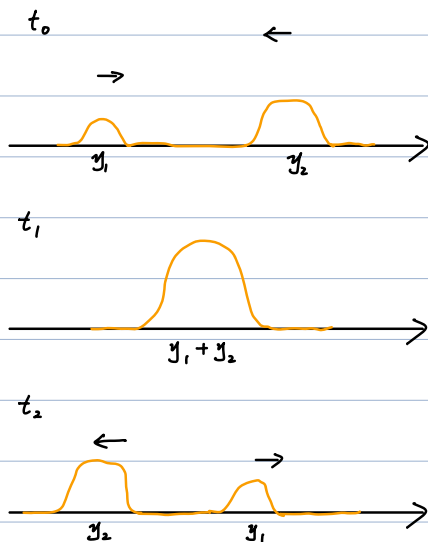
$$y = f(x + vt)$$

v : wave speed (波速、相位速度)

Principle of superposition:

$$y'(x, t) = y_1(x, t) + y_2(x, t)$$

Overlapping waves add up algebraically.

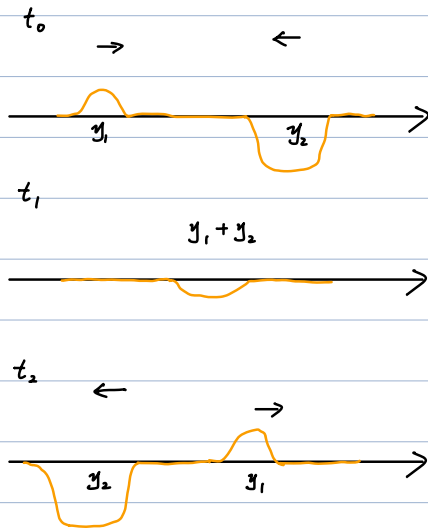


Linear waves: waves that obey superposition principle.

Interference: the phenomenon of combining waves.

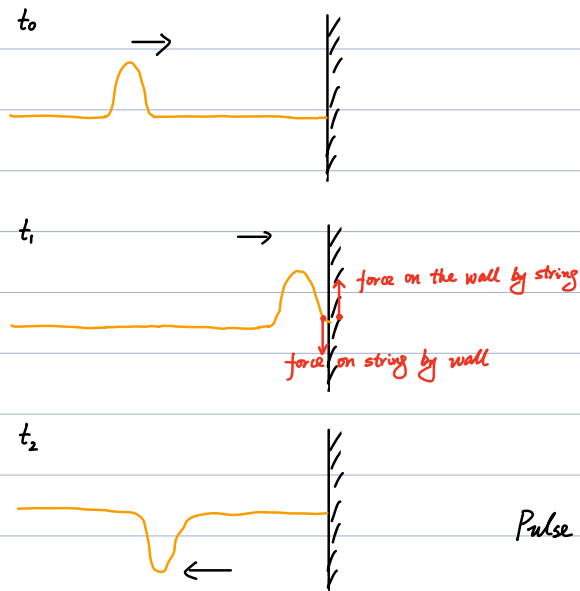
Previous example: Constructive interference. (相长干涉、建设性干涉)

Destructive interference (相消干涉、破坏性干涉)



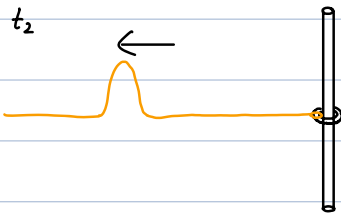
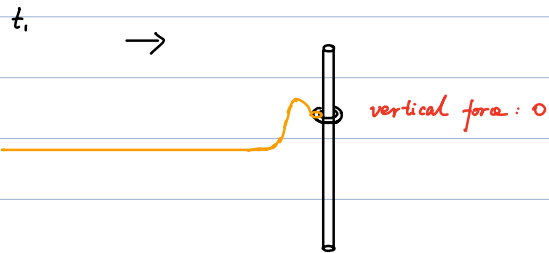
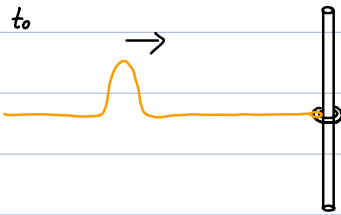
Reflection (upon change of medium)

fixed boundary.

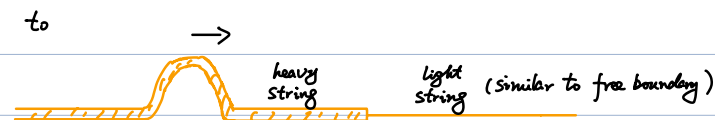
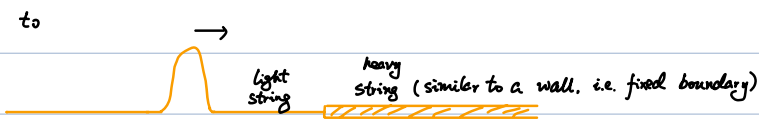


Pulse inverts upon reflection.

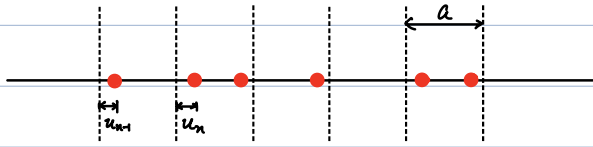
free boundary



Transmission



Example. mechanical waves in monoatomic crystal (phonon)



Equilibrium: $X_n = na$

Deviation: $u_n = x_n - X_n$

Nearest neighbor interaction (potential energy):

$$\begin{aligned}\phi(x_{n+1} - x_n) &= \phi_0 + \frac{1}{2} K (x_{n+1} - x_n - a)^2 + \dots \\ &= \phi_0 + \frac{1}{2} K (u_{n+1} - u_n)^2 + \dots\end{aligned}$$

$$\Rightarrow U^{\text{total}} \approx \frac{K}{2} \sum_n (u_{n+1} - u_n)^2$$

$$F_n = - \frac{dU^{\text{total}}}{du_n} = m \ddot{u}_n$$

$$\Rightarrow m \ddot{u}_n = K(u_{n+1} - u_n) - K(u_n - u_{n-1})$$

$$= Ka \frac{u_{n+1} - u_n}{a} - Ka \frac{u_n - u_{n-1}}{a}$$

$$\stackrel{\lim_{a \rightarrow 0}}{=} Ka \left. \frac{\partial u}{\partial x} \right|_{x_n + \frac{a}{2}} - Ka \left. \frac{\partial u}{\partial x} \right|_{x_n - \frac{a}{2}}$$

$$= Ka^2 \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_n}$$

$$\Rightarrow \boxed{\frac{\partial^2 u}{\partial t^2} = v^2 \frac{\partial^2 u}{\partial x^2}} \quad v \equiv a \sqrt{\frac{K}{m}}$$

Linear wave equation $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$ applies to various waves:

- Sound wave, y : displacement of air molecule
- wave on the string, y : vertical displacement
- E&M. y : electric or magnetic field.
- ...

Note:

Physical meaning of "taking the continuum limit", i.e., $\lim_{a \rightarrow 0}$:

$\lambda \gg a$ (wave length much bigger than lattice spacing)

or equivalently, $k \ll \frac{2\pi}{a}$ (see sinusoidal wave below)

Example. wave on string (assuming small amplitude)

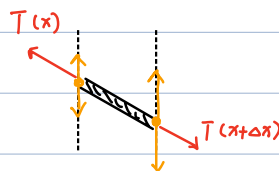
$$\mu \Delta x \frac{\partial^2 y}{\partial t^2} = (-T(x+\Delta x)) \cdot \left| \frac{\partial y}{\partial x} \right|_{x+\Delta x} + T(x) \cdot \left| \frac{\partial y}{\partial x} \right|_x$$

$\downarrow$
 mass per length

$$\approx T \Delta x \frac{\partial^2 y}{\partial x^2} \Big|_x$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} \equiv v^2 \frac{\partial^2 y}{\partial x^2}$$

$$\text{where } v \equiv \sqrt{\frac{T}{\mu}}$$



Obvious solutions to linear wave equation:

$$y = f(x - vt) \quad \text{right-moving}$$

$$y = f(x + vt) \quad \text{left-moving}$$

check $f(x - vt)$:

$$\frac{\partial y}{\partial t} = -v f'(x - vt)$$

$$\frac{\partial^2 y}{\partial t^2} = v^2 f''(x - vt)$$

$$\frac{\partial y}{\partial x} = f'(x - vt)$$

$$\frac{\partial^2 y}{\partial x^2} = f''(x - vt)$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad \checkmark$$